Math Practice Test J

1 A recipe uses 3 cups of flour for every 2 cups of milk. How much flour goes with 8 cups of milk?

Working

Answer

2 2 pens cost $3, 4 pens cost $6, and 6 pens cost $10. Is the cost proportional to the number of pens? Write yes or no.

Working

Answer

3 A taxi charges $3 plus $2 for each mile. A ride costs $17. How many miles was the ride?

Working

Answer

4 A triangular garden bed has a base of 8 m and a height of 5 m. What is its area?

Working

Answer

5 Work out 4.6×10^3

Working

Answer

6 Without working it out: is $5/6 \times 72$ greater than or less than 72?

Working

Answer

7 The 7 in 0.7 is worth how many times the 7 in 0.07?

Working

Answer

8 Work out $(-8) \times 3/4 \times (-1/2)$

Working

Answer

9 A ribbon 1 3/4 m long is joined to one 2 1/3 m long. What is the total length?

Working

Answer

10 Start at the origin. Move 5 to the right and 3 up. Write the coordinates of the point.

Working

Answer

Answer Key: Math Practice Test J

Answers involving money or ages should include units ($, ¢, years). Working shows one way to solve each.

Q	Answer	Working	Skill checked
1	**12 cups**	8 is 4 lots of 2, so 4 lots of 3 = 12 cups	Ratio reasoning in a real-world problem (6.RP.A.3)
2	**no**	Price per pen: $1.50, $1.50, then $1.67 - not the same, so not proportional	Recognize a proportional relationship (7.RP.A.2)
3	**7 miles**	3 + 2m = 17, so 2m = 14 and m = 7	Write and solve an equation from a word problem (7.EE.B.4)
4	**20 m²**	Half of base × height: (8 × 5) ÷ 2 = 20, in square meters	Area of a triangle in a real-world problem (6.G.A.1)
5	**4600**	$10^3 = 10 \times 10 \times 10 = 1000$, so the decimal point moves three places right	Multiply a decimal by a power of 10 (5.NBT.A.2)
6	**less than**	5/6 is less than 1, so it shrinks 72	Multiplication as scaling (5.NF.B.5)
7	**10**	0.7 ÷ 0.07 = 10: one place to the left is worth 10 times as much	A digit is worth 10 times the place to its right (5.NBT.A.1)
8	**3**	Two negatives give a positive. 8 × 1/2 = 4, then 4 × 3/4 = 3	Multiply rational numbers using properties of operations (7.NS.A.2.c)
9	**4 1/12 m**	Twelfths: 1 9/12 + 2 4/12 = 3 13/12 = 4 1/12 m	Fraction addition word problem (5.NF.A.2)
10	**(5, 3)**	Across 5 first, then up 3	Interpret both coordinates as moves from the origin (5.G.A.1)